

Theorema Egregium

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April 18, 2026

Abstract

Following Gauss's footsteps, we develop a method for quantifying the curvature of 2-dimensional surfaces, then show that this notion of curvature can be used as a "compatibility condition" when determining whether or not one surface can serve as a faithful cartographic map of another—i.e., one which preserves angles, distances (up to a constant scale factor k), and areas (up to a scale factor k^2). We then apply this compatibility condition to the analysis of two real-world problems: (1) mapping the surface of the Earth and (2) optimal pizza-wielding technique.

1 Basic Definitions

A **parametric surface** is defined by a smooth function $\vec{r} : D \rightarrow \mathbb{R}^3$ where D is an open, connected subset of $\mathbb{R}^2$. D acts as the coordinate space used to parameterize the surface, while $\vec{r}$ maps each coordinate pair $(u_1, u_2) \in D$ to the point in $\mathbb{R}^3$ representing the physical location of the point on the surface described by those coordinates. The surface S is then formed as the image of the entire coordinate space under the map $\vec{r}$:

$$S := \vec{r}(D) := \{\vec{r}(u_1, u_2) \mid (u_1, u_2) \in D\}$$

We will additionally assume that $\vec{r}$ is **regular**—the partial derivatives of $\vec{r}$ with respect to its first and second arguments (resp. $\partial_1 \vec{r}$ and $\partial_2 \vec{r}$) are linearly independent whenever they are evaluated at the same point $(u_1, u_2) \in D$, for all points in D . This guarantees that, at every point on the surface, varying one coordinate results in motion which cannot be achieved by varying the other coordinate. The smoothness and regularity of $\vec{r}$ together justify the intuition that S is a 2-dimensional object embedded in 3-dimensional space.

Given a pair of smooth functions $u_1, u_2 : I \rightarrow \mathbb{R}$ where I is an open, non-empty interval of $\mathbb{R}$, we can define a curve on S given by:

$$\begin{aligned} \vec{\gamma} : I &\rightarrow \mathbb{R}^3 \\ t &\mapsto \vec{r}(u_1(t), u_2(t)) \end{aligned}$$

Each function $u_i(t)$ can be interpreted as indicating the value of the i^{th} coordinate at time t . The function $\vec{\gamma}$ then uses these coordinate functions to define a curve in $\mathbb{R}^3$ which is contained in S . The velocity of such a curve at time t is given by:

$$\dot{\vec{\gamma}}(t) = \dot{u}_1(t)[\partial_1 \vec{r}](u_1(t), u_2(t)) + \dot{u}_2(t)[\partial_2 \vec{r}](u_1(t), u_2(t))$$

Where $\dot{u}_i$ is the i^{th} component of the **coordinate velocity**—the rate of change of the coordinate tuple $(u_1(t), u_2(t))$ with respect to t . Since the velocity $\dot{\vec{\gamma}}(t)$ is tangent to the curve at the point $\vec{\gamma}(t)$, and $\vec{\gamma}$ is fully contained in S by construction, $\dot{\vec{\gamma}}(t)$ is therefore also tangent to the surface S at the point $\vec{\gamma}(t)$. This motivates the definition of the **tangent plane** $T_{\vec{r}(u_1, u_2)}S$ at the point $\vec{r}(u_1, u_2)$ for each $(u_1, u_2) \in D$ as the set of vectors given by:

$$T_{\vec{r}(u_1, u_2)}S := \{\alpha[\partial_1 \vec{r}](u_1, u_2) + \beta[\partial_2 \vec{r}](u_1, u_2) \mid (\alpha, \beta) \in \mathbb{R}^2\}$$

This plane can be understood as the set of all velocity vectors associated with all possible curves on S passing through the point $\vec{r}(u_1, u_2)$. It is evident that this is a 2-dimensional subspace of $\mathbb{R}^3$ since it is defined as the linear span of the vectors $[\partial_1 \vec{r}](u_1, u_2)$ and $[\partial_2 \vec{r}](u_1, u_2)$.

Additionally, we define the **surface normal** $\vec{N}(u_1, u_2)$ and **unit surface normal** $\hat{n}(u_1, u_2)$ vectors at $\vec{r}(u_1, u_2)$ as:

$$\begin{aligned}\vec{N}(u_1, u_2) &:= [\partial_1 \vec{r}](u_1, u_2) \times [\partial_2 \vec{r}](u_1, u_2) \\ \hat{n}(u_1, u_2) &:= \frac{\vec{N}(u_1, u_2)}{|\vec{N}(u_1, u_2)|}\end{aligned}$$

These vectors stand orthogonal to $T_{\vec{r}(u_1, u_2)}S$ (and therefore also S itself) at the point $\vec{r}(u_1, u_2)$ and are guaranteed to be non-zero everywhere on S due to the regularity of $\vec{r}$. Note that the set $\{[\partial_1 \vec{r}](u_1, u_2), [\partial_2 \vec{r}](u_1, u_2), \hat{n}(u_1, u_2)\}$ is a linearly independent set of three vectors in $\mathbb{R}^3$, so any vector in $\mathbb{R}^3$ can be decomposed as a unique linear combination of them. The same is true if $\vec{N}(u_1, u_2)$ is substituted for $\hat{n}(u_1, u_2)$.

The unit surface normal $\hat{n}$ can additionally be interpreted as a map from D to the unit sphere $\mathbb{S}^2$ (*not* the Cartesian product of the surface S with itself) due to its output always being unit-length. Under this interpretation, it is known as the **Gauss map**.

2 Notation

The functions $\vec{r}$, $\vec{N}$, $\hat{n}$, and all of their partial derivatives are implicitly evaluated at the same point $(u_1, u_2) \in D$ whenever they appear in an expression without being given explicit arguments, so that they are shorthands for the $\mathbb{R}^3$ vectors resulting from those evaluations. Any statement made about these vectors implicitly carries the qualifier “for all $(u_1, u_2) \in D$ ”.

3 What exactly do we mean by “curvature”?

We begin our exploration of surface curvature by investigating, for a given patch of coordinate space $E \subseteq D$, the relationship between the signed area $A_{\vec{r}(E)}$ on S covered by $\vec{r}(E)$ and the signed area $A_{\hat{n}(E)}$ on $\mathbb{S}^2$ covered by $\hat{n}(E)$:

$$\begin{aligned}A_{\vec{r}(E)} &= \int_E (\partial_1 \vec{r} \times \partial_2 \vec{r}) \cdot \hat{n} \, du_1 \, du_2 \\ A_{\hat{n}(E)} &= \int_E (\partial_1 \hat{n} \times \partial_2 \hat{n}) \cdot \hat{n} \, du_1 \, du_2\end{aligned}$$

The motivation for doing so is this: $\mathbb{S}^2$ effectively represents the “space of directions” one can face in $\mathbb{R}^3$. The area $A_{\hat{n}(E)}$ then represents the portion of this “space of directions” covered by the unit surface normals of $\vec{r}(E)$. This suggests a definition of curvature based on “amount of direction space covered per unit surface area,” which directs our attention to the relationship between the integrands.

First, we analyze the partial derivatives $\partial_i \hat{n}$:

$$\begin{aligned}\partial_i \hat{n} &= \partial_i \left(\frac{\vec{N}}{|\vec{N}|} \right) \\ &= \frac{1}{|\vec{N}|^2} (|\vec{N}| \partial_i \vec{N} - (\partial_i |\vec{N}|) \vec{N}) \\ &= \frac{1}{|\vec{N}|^2} (|\vec{N}| \partial_i \vec{N} - (\partial_i ((\vec{N} \cdot \vec{N})^{\frac{1}{2}})) \vec{N}) \\ &= \frac{1}{|\vec{N}|^2} \left(|\vec{N}| \partial_i \vec{N} - \left(\frac{1}{2} (\vec{N} \cdot \vec{N})^{-\frac{1}{2}} (2(\partial_i \vec{N}) \cdot \vec{N}) \right) \vec{N} \right) \\ &= \frac{1}{|\vec{N}|^2} \left(|\vec{N}| \partial_i \vec{N} - \left(\frac{1}{|\vec{N}|} (\partial_i \vec{N}) \cdot \vec{N} \right) \vec{N} \right) \\ &= \frac{1}{|\vec{N}|} (\partial_i \vec{N} - ((\partial_i \vec{N}) \cdot \hat{n}) \hat{n})\end{aligned}$$

The expression $(\partial_i \vec{N} - ((\partial_i \vec{N}) \cdot \hat{n})\hat{n})$ subtracts from $\partial_i \vec{N}$ the component of itself which is parallel to $\hat{n}$, leaving only the orthogonal component which we denote by $(\partial_i \vec{N})_{\perp \hat{n}}$. $(\partial_i \vec{N})_{\perp \hat{n}}$ is therefore an element of the tangent plane $T_{\vec{r}}S$, so there exist coefficients α and β such that:

$$\partial_i \hat{n} = (\partial_i \vec{N})_{\perp \hat{n}} = \alpha(\partial_1 \vec{r}) + \beta(\partial_2 \vec{r})$$

To find these coefficients, we first define the following values based on dot products formed using $\partial_1 \vec{r}$ and $\partial_2 \vec{r}$:

$$E = \partial_1 \vec{r} \cdot \partial_1 \vec{r} \tag{1}$$

$$F = \partial_1 \vec{r} \cdot \partial_2 \vec{r} \tag{2}$$

$$G = \partial_2 \vec{r} \cdot \partial_2 \vec{r} \tag{3}$$

We then have:

$$(\partial_i \vec{N})_{\perp \hat{n}} \cdot \partial_1 \vec{r} = \alpha E + \beta F \tag{4}$$

$$(\partial_i \vec{N})_{\perp \hat{n}} \cdot \partial_2 \vec{r} = \alpha F + \beta G \tag{5}$$

We can cancel all terms involving β by multiplying Eq. (4) by G , multiplying Eq. (5) by F , then subtracting the two. Rearranging to isolate α gives:

$$\alpha = \frac{((\partial_i \vec{N})_{\perp \hat{n}} \cdot \partial_1 \vec{r})G - ((\partial_i \vec{N})_{\perp \hat{n}} \cdot \partial_2 \vec{r})F}{EG - F^2} \tag{6}$$

A similar manipulation can be used to solve for β , giving:

$$\beta = \frac{((\partial_i \vec{N})_{\perp \hat{n}} \cdot \partial_2 \vec{r})E - ((\partial_i \vec{N})_{\perp \hat{n}} \cdot \partial_1 \vec{r})F}{EG - F^2} \tag{7}$$

Note that the assumed linear independence of $\partial_1 \vec{r}$ and $\partial_2 \vec{r}$ guarantees that:

$$EG - F^2 = (\partial_1 \vec{r} \cdot \partial_1 \vec{r})(\partial_2 \vec{r} \cdot \partial_2 \vec{r}) - (\partial_1 \vec{r} \cdot \partial_2 \vec{r})^2 > 0$$

4 The First Fundamental Form, I

In the standard literature, the quantities E , F , and G in the preceding section are grouped into the following matrix called the **first fundamental form**, whose determinant appears in the denominators of Eq. (6) and Eq. (7).

$$\mathbf{I} = \begin{bmatrix} E & F \\ F & G \end{bmatrix} \tag{8}$$

The first fundamental form answers the following geometric question: suppose we are given two tangent vectors $\vec{v}_1, \vec{v}_2 \in T_{\vec{r}}S$ in terms of their components along $\partial_1 \vec{r}$ and $\partial_2 \vec{r}$; what is the dot product of $\vec{v}_1$ and $\vec{v}_2$, considered as elements of $\mathbb{R}^3$?

Let:

$$\vec{v}_1 = \alpha_1 \partial_1 \vec{r} + \beta_1 \partial_2 \vec{r}$$

$$\vec{v}_2 = \alpha_2 \partial_1 \vec{r} + \beta_2 \partial_2 \vec{r}$$

Then:

$$\begin{aligned} \vec{v}_1 \cdot \vec{v}_2 &= (\alpha_1 \partial_1 \vec{r} + \beta_1 \partial_2 \vec{r}) \cdot (\alpha_2 \partial_1 \vec{r} + \beta_2 \partial_2 \vec{r}) \\ &= \alpha_1 \alpha_2 E + (\alpha_1 \beta_2 + \beta_1 \alpha_2) F + \beta_1 \beta_2 G \\ &= \begin{bmatrix} \alpha_1 & \beta_1 \end{bmatrix} \mathbf{I} \begin{bmatrix} \alpha_2 \\ \beta_2 \end{bmatrix} \end{aligned}$$

That is, the first fundamental form allows us to compute the physical-space ($\mathbb{R}^3$) dot product of the coordinate velocity vectors $\vec{v}_1, \vec{v}_2 \in T_{\vec{r}}S$ in terms of their respective coordinate-space representations $(\alpha_1, \beta_1), (\alpha_2, \beta_2) \in \mathbb{R}^2$. I therefore capture critical information about the coordinate-space-to-physical-space mapping defined by $\vec{r}$.

5 Appendices

6 Shelf

$$\begin{aligned}
dA_{\vec{n}(E)} &= (\partial_1 \hat{n} \times \partial_2 \hat{n}) \cdot \hat{n} \\
&= (\partial_1 (\vec{N}/|\vec{N}|) \times \partial_2 (\vec{N}/|\vec{N}|)) \cdot \hat{n} \\
&= \frac{1}{|\vec{N}|^2} [(|\vec{N}| \partial_1 \vec{N} - (\partial_1 |\vec{N}|) \vec{N}) \times (|\vec{N}| \partial_2 \vec{N} - (\partial_2 |\vec{N}|) \vec{N})] \cdot \hat{n} \\
&= \frac{1}{|\vec{N}|^2} [|\vec{N}|^2 (\partial_1 \vec{N} \times \partial_2 \vec{N}) - |\vec{N}| (\partial_1 |\vec{N}|) (\vec{N} \times \partial_2 \vec{N}) - |\vec{N}| (\partial_2 |\vec{N}|) (\partial_1 \vec{N} \times \vec{N})] \cdot \hat{n} \\
&= (\partial_1 \vec{N} \times \partial_2 \vec{N}) \cdot \hat{n}
\end{aligned}$$